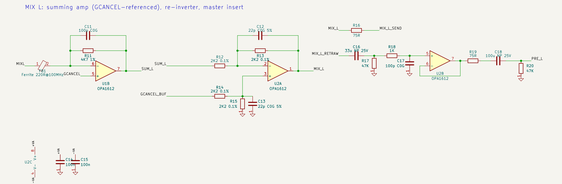
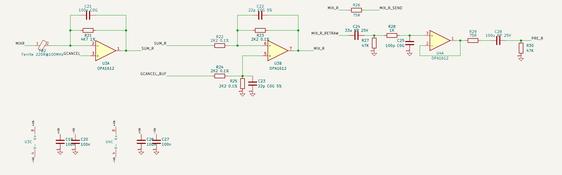
GROUND CANCEL BUFFER (feeds every re-inverter; summing amps use the raw GCANCEL line; 100K keeper)



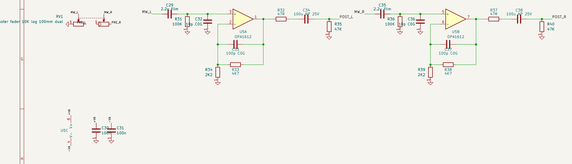
MIX L: summing amp (GCANCEL-referenced), re-inverter, master insert



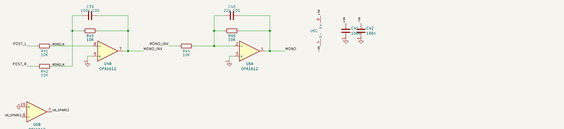
MIX R: summing amp (GCANCEL-referenced), re-inverter, master insert



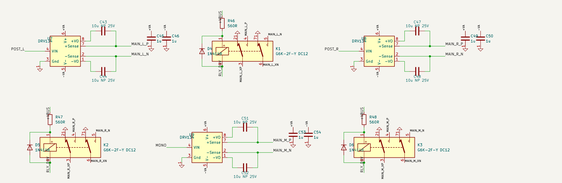
MASTER FADER (dual 100 mm A10K, pin 1 = GNDA) + post-amps +10 dB



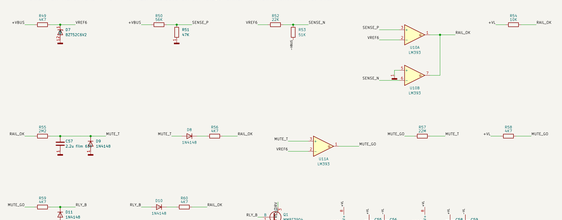
MONO: inverting sum of POST L+R, re-inverter



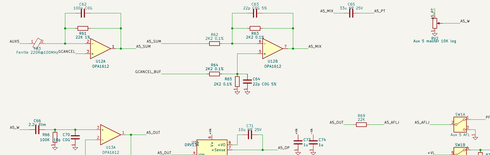
MAIN OUTPUTS: DRV134 -> 10u NP in the Sense loop -> mute relay (NO: NC grounds the line) -> XLR male, pin 2 hot



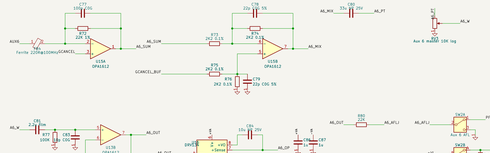
POWER-ON MUTE: rails > -14 V for ~2.7 s closes all six relays; either rail low opens them at once



AUX MASTER 5: OPA1612 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL

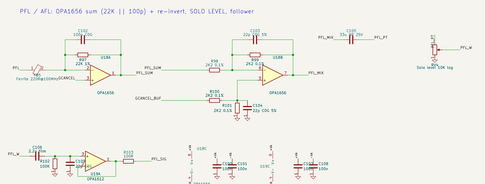


AUX MASTER 6: OPA1612 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL

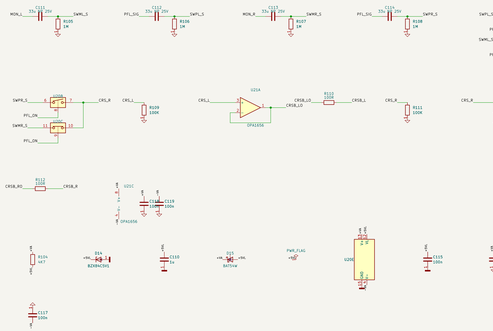


EXT INPUT (2-track / DAW) and MIX / EXT monitor select (SW latched = EXT)

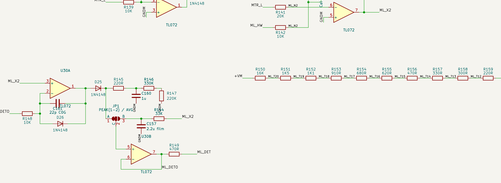
PFL / AFL: OPA1656 sum (22K || 100p) + re-invert, SOLO LEVEL, follower



PFL SWITCHER: DG413 (PFL pressed = SW1/SW4, else SW2/SW3; break-before-make) -> CR5 L/R OPA1656 followers



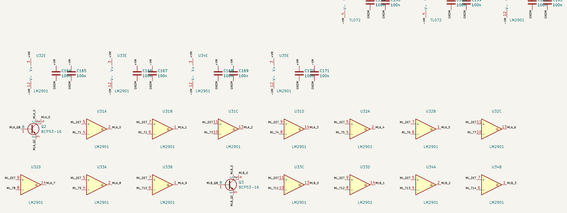
PFL DETECT (0.92 V per press vs 0.50 V: 2 x 1N4148 clamp, 1u debounce) and DG413 logic, beside the switcher



RUDE SOLO: ~2 Hz flasher, PFL ACTIVE LED; held off through FL_N while no key is pressed



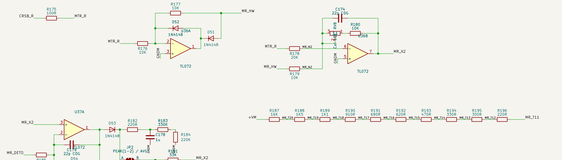
CONTROL ROOM: C/ROOM dual A10K, followers, DRV134, mute relays, TRS outs



HEADPHONES: PHONES dual A10K, OPA1612 + LME49600 composite (+9.9 dB); 2R2 out, mute relay, PHONESGND

METER L: full-wave rectifier, precision peak / average, follower (TL072); 2 dB ladder, 2 x 10-LED strings

METER R: full-wave rectifier, precision peak / average, follower (TL072); 2 dB ladder, 2 x 10-LED strings



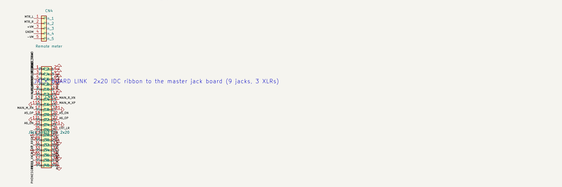
POWER ENTRY (7-pin XLR NC7MD-LX on the rear panel, two 5 A contacts per rail, 18 AWG to this terminal; +-17 V PSU)

DISTRIBUTION (PTC per wing per rail, 1.85 A hold, 72 V; local branches VA audio, VP phones, VM meters, VL logic)

MOTHERLOOM (master in the centre: CN1 = left wing, CN2 = right wing; Delta pinout)



REMOTE METER HEADER (for a meter bridge)



Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9

Sheet: /
File: master.kicad_sch

Title: Master section — goa console

Size: User		Date: 2026-05-19	Rev: 0.9
KiCad 8.0.x 10.0.2			Id: 1/1